Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2602903

Dancing with Data: Latent Vote Inference and Fairness via TRAC-MH and GDSTree

Summary

To address the "black-box" ambiguity of fan votes in *Dancing with the Stars*, we developed a comprehensive Inverse Reconstruction & Mechanism Design Framework. This system inferred latent preferences, evaluated aggregation biases, isolated performance drivers, and proposed fairer scoring protocols to resolve popularity–merit conflicts.

Firstly, we reconstructed latent fan votes using the **Temporal Recursive Adaptive Constrained Metropolis–Hastings** framework. Integrating our innovative time-dependent Dirichlet prior with a stochastic mixture kernel, this Bayesian model achieved high accuracy in weekly survival margins and in winner prediction. To increase global consistency, we applied **GDSTree**, using lookahead simulation to boost feasible-path consistency.

Secondly, sensitivity analysis confirmed that the Percentage method was the most fan-favoring, yielding higher Spearman's rank correlation coefficient and lower mean absolute difference in rankings than Rank or Save. Counterfactual tests showed that Rank vs. Percentage rarely changed outcomes under overwhelming fan support, while Judges' Save could remove extreme low-performance outliers.

Thirdly, we decoupled performance drivers using **the XGBoost – SHAP Model**. By decomposing contributions across five attributes, we established a clear hierarchy: Age was the dominant determinant, showing a sharp negative effect for older contestants. Partner assignment served as a strong secondary driver, whereas Homestate and Industry impacted results primarily through outlier fanbases.

Finally, we proposed two optimization mechanisms. **Base Vote Smoothing Method** compresses extreme vote gaps, improving fairness while maintaining high fan alignment. **Dynamic Adaptive Weighting Method** shifts weights from fan-centric to judge-centric as the season progressed, balancing engagement in the early stage with professional integrity in the finale. Sensitivity analysis confirmed our model's robustness under parameter variations. Overall, our framework offered actionable insights for fair and engaging mechanism design.

Keywords: TRAC-MH Framework; GDSTree; XGBoost; SHAP Analysis; Mechanism Design; Sensitivity Analysis

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1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) operates on a dual-input system where celebrity couples receive technical scores from judges and "save" fan votes from the audience. The competition's core mechanic involves aggregating these distinct inputs into a single metric, where the couple with the lowest combined score is eliminated each week. Historically, the aggregation logic has shifted between two primary methods: Rank Sums (adding ordinal placements) and Percentage Sums (adding proportional shares). Recently, to mitigate controversies where popularity overshadowed low technical scores, the show reintroduced Rank-based aggregation supplemented by a "Judges' Save," allowing judges to rescue one of the bottom two couples.

Given the confidentiality of specific fan vote totals, we must construct a model to infer these latent values from observable judge scores and elimination results. Utilizing this reconstructed dataset, we investigate the determinants of fan preference, quantify the trade-offs between technical merit and viewer popularity, and assess the systematic impact of different aggregation mechanisms.

1.2 Restatement of the Problem

To reconcile the discrepancy between fan votes and judge scores for a fairer ranking system, we frame our approach through four distinct mathematical modeling tasks:

- **Task 1: Reconstruction of Latent Fan Vote Distributions (The Inverse Problem).** The primary objective is to reverse-engineer the unobserved audience voting preferences. By treating fan votes as latent variables constrained by historical elimination outcomes, we aim to derive their posterior probability distributions, thereby quantifying the uncertainty and consistency of these hidden inputs.
- **Task 2: Sensitivity Analysis of Aggregation Mechanisms.** With the reconstructed data, we simulate the competition under alternative protocols (Rank-based vs. Percentage-based). This task evaluates the system's sensitivity to aggregation logic, specifically identifying structural biases that may disproportionately amplify popularity over technical merit.
- **Task 3: Factor Decomposition of Competitive Performance.** We seek to decouple the marginal effects of static attributes (e.g., celebrity demographics, professional partnerships) from week-to-week performance. Our goal is to statistically isolate the drivers of judge scores (technical quality) versus fan support (public appeal).
- **Task 4: Mechanism Design for Optimal Fairness.** Based on the identified biases, we design an optimized scoring framework. This involves proposing a new aggregation function that maximizes a defined fairness metric while maintaining competitive viability, culminating in a strategic policy memo.

1.3 Literature Review

We synthesize four types of literature to bridge gaps in temporal reconstruction and fairness analysis.

Dynamic Bayesian Modeling. Reconstructing hidden votes requires handling temporal inertia and strict constraints. Building on West et al.'s recursive Bayesian updating [1] and Khormi et al.'s indicator-based inference [2], our TRAC-MH framework synthesizes these approaches to track vote momentum while strictly enforcing binary elimination outcomes.

Sequential Optimization. To avoid the "dead-end" trajectories common in myopic sampling, we leverage GDSTree. Following Browne et al.'s framework for sequential decision processes [3], we implement a VoteOptimizer that uses lookahead simulations to prune infeasible paths, ensuring global consistency across the season.

Social Influence and Fairness. Theory suggests that social influence can impact collective decision-making. Danchin et al. and Lorenz et al. illustrate how public signals can affect individual judgment, highlighting the complex interactions between popularity and performance [4, 5]. These insights inform our analysis in Task 2, supporting the implementation of a "Judges' Save" mechanism to address instances when popularity diverges from technical merit, ensuring a balanced assessment of participants.

Robust Factor Analysis. To quantify non-linear performance drivers, we adopt XGBoost [6]. Its gradient-boosted structure with explicit regularization outperforms linear models in handling noisy voting data, allowing us to robustly isolate the marginal effects of attributes like Age and Partnership in Task 3.

1.4 Overview of Our Work

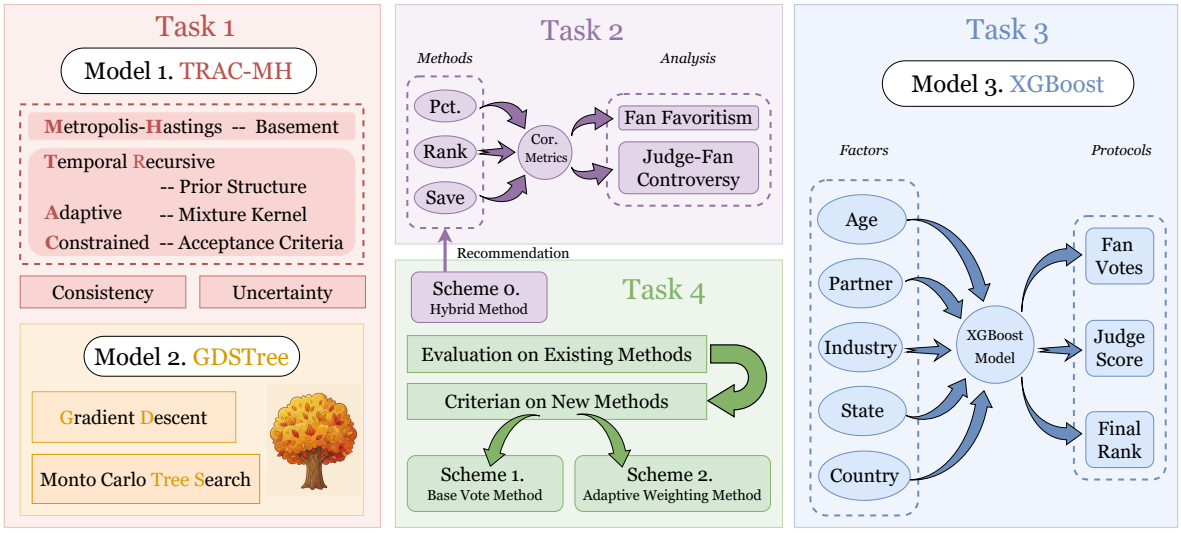


Figure 1: Schematic overview of the proposed research framework. The workflow bridges data reconstruction (Task 1), sensitivity and factor analysis (Tasks 2 & 3), and mechanism optimization (Task 4).

2 Preparations

To simplify the complex dynamics of voting behavior and ensure the mathematical tractability of our model, we propose the following fundamental assumptions:

Assumption 1: Initial Neutrality of Audience Preference
Explanation: We assume that at the start of a new season, the audience has no prior preference bias toward any contestant, as defined in (1). This "Principle of Indifference" provides a neutral starting point and simplifies the initial Bayesian.

Assumption 2: Persistence of Fan Support
Explanation: We assume that a contestant's fan share Φ_t remains proximal to their estimated performance from the previous week $\hat{\phi}_{t-1}$.

Assumption 3: Conditional Independence of Judge Scores
Explanation: We assume that judge scores are independent of fan votes(i.e. $P(\Phi_t|J_t) = P(\Phi_t)$). This simplifies the posterior to Eq. (3).

Assumption 4: Consistency of Judge Scores
Explanation: We assume that judges' preference remain stable within the same week. Consequently, judges will save the contestant with the higher Judge rank in the bottom two. This assumption is reasonable and facilitates the analysis regard to "Judges' Save" mechanism.

We summarize the main notations and variables in Table 1. Definitions are included to avoid ambiguity across tasks.

Table 1: Symbols and Definitions

Symbol	Definition
s, t, i	Indices for season, week, and contestant, respectively
N_t	Number of active contestants in week t
$J_{i,t}$	Judge's technical score for contestant i in week t
$E_{i,t}$	Binary elimination status ($1 =$ Eliminated, $0 =$ Safe)
Φ_t	Latent fan vote distribution vector at week t
$\Delta_{s,t}$	Safety Margin: Gap between survivor min. and eliminated max.
U_{SD}	Posterior SD: Metric for estimation uncertainty
m	Aggregation method index (R: Rank, P: Percentage, S: Save)
$r_{i,t}^{(m)}$	Final rank calculated under aggregation method m
$\rho_{s,t}^{(m)}$	Spearman correlation between method m and fan preferences
$\text{MAD}^{(m)}$	Mean Absolute Deviation between method m and fan ranks
NRI	Normalized Rank Inversion: Measure of judge-fan disagreement
Δp	Percent Gap: Difference between fan vote and judge score shares
ϕ_j^{SHAP}	SHAP value contribution of feature j (e.g., Age)
W_{fan}	Dynamic fan vote weight in the proposed mechanism

3 Task 1: The TRAC-MH Reconstruction Model

To address the challenge of reconstructing latent fan preferences from partial elimination data, we propose the **TRAC-MH** framework (**T**emporal **R**ecursive **A**daptive **C**onstrained **M**etropolis-**H**astings), which was implemented in Python [1].

This probabilistic model inverts the elimination mechanism to infer the unobserved vote distribution Φ_t. The framework is structured into four systematic stages. **First**, we standardize the raw contest data to establish a rigorous computational foundation. **Second**, the core inference engine combines a time-dependent prior with a specialized sampler to navigate the constrained solution space (Sections 3.1–3.2). **Third**, we validate model reliability through historical consistency checks and uncertainty quantification (Sections 3.3–3.4). **Finally**, we extend the framework using Monte Carlo Tree Search to optimize sequential decision paths (Section 3.5).

To enable modeling, we transformed the raw dataset into a standardized "long" format. A critical component of this pipeline is our Conservative Labeling Strategy: voluntary withdrawals are coded as "Safe" ($E_{i,t} = 0$) rather than eliminations ($E_{i,t} = 1$). This ensures that the model's hard constraints are only triggered by actual popularity-based outcomes, preventing bias from non-competitive exits.

3.1 Time-Dependent Prior Construction

To capture the "temporal stickiness" of fan support, we reject static models in favor of a Recursive Bayesian System, where past posterior knowledge informs the present prior [1].

Initialization and Propagation

For the first week ($t = 1$), applying the Principle of Indifference, we adopt a non-informative symmetric Dirichlet distribution as the prior for the fan vote vector Φ_1:

$$P(\Phi_1) \sim \text{Dirichlet}(\alpha_1), \quad \text{where } \alpha_1 = (1, 1, \ldots, 1) \tag{1}$$

This represents a uniform distribution over the simplex. For subsequent weeks ($t > 1$), the estimated posterior mean from the previous week ($\hat{\phi}_{i,t-1}$) propagates to construct the current prior parameter $\alpha_{i,t}$:

$$\alpha_{i,t} = \lambda \cdot \hat{\phi}_{i,t-1} + \varepsilon \tag{2}$$

where λ (set to ≈ 50) is a concentration hyperparameter controlling the strength of historical momentum, and ε is a smoothing factor.

Dynamic Alignment Mechanism

Since the contestant pool shrinks weekly, we implement a three-stage alignment to construct the prior for week t:

- **Dimension Reduction:** The vector space is reduced from $\mathbb{R}^N$ to $\mathbb{R}^{N-1}$ by simply discarding dimensions corresponding to eliminated contestants.

[1] During the implementation, we utilized AI-assisted coding tools strictly for syntax auto-completion and library function verification [7, 8]. No core algorithm logic was generated by AI.

- **Parameter Inheritance:** Surviving contestants directly inherit their α parameters (as defined in Eq. (2). This preserves the "popularity momentum" accumulated in historical data without distortion.
- **Implicit Re-normalization:** By feeding the subset of surviving α parameters into the Dirichlet distribution, the model automatically re-normalizes the probability mass. This ensures that the vote share of eliminated contestants ("orphan votes") is proportionally re-distributed among the survivors.

3.2 The TRAC-MH Framework

We developed the TRAC-MH Framework (Temporal Recursive Adaptive Constrained Metropolis-Hastings) to reconstruct the latent fan vote distribution Φ_t. By sampling strictly within the feasible region, this framework ensures all generated vectors are consistent with historical elimination outcomes.

The comprehensive workflow of the framework is visualized in Figure 2.

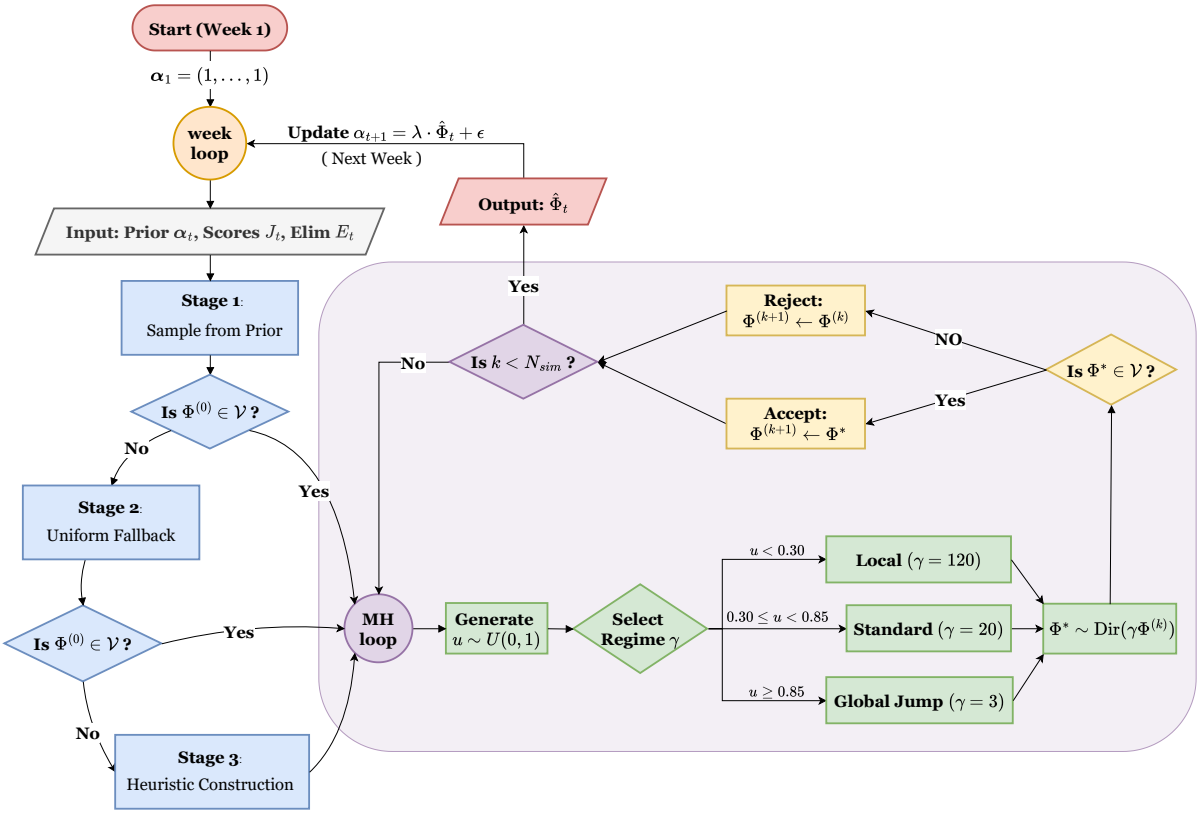


Figure 2: Workflow of the TRAC-MH Framework (in one season).

Bayesian Formulation

Our goal is to sample from the posterior distribution $P(\Phi_t \mid J_t, E_t)$. By applying Bayes' Theorem and assuming judge scores are independent of fan votes, the normalization constant becomes irrelevant for sampling. The target density is thus derived as:

$$P(\Phi_t|J_t,E_t) = \frac{P(E_t|J_t,\Phi_t)\cdot P(J_t|\Phi_t)\cdot P(\Phi_t)}{P(E_t,J_t)} \propto \underbrace{P(E_t|J_t,\Phi_t)}_{\text{Likelihood } \mathscr{L}(\Phi)} \cdot \underbrace{P(\Phi_t)}_{\text{Prior } \pi(\Phi)} \tag{3}$$

- **The likelihood term** $P(E_t|J_t,\Phi_t)$ acts as a binary indicator function, assigning a probability of 1 to feasible vectors and 0 to impossible ones.
- **The prior term** $P(\Phi_t)$ is defined primarily through the time-series approach described in Section 3.1. Specific operational details are elaborated in the immediately following Section 3.2.1.

3.2.1 Hierarchical Initialization Strategy

To instantiate the MH sampler for the target density defined above, we require a valid starting point $\Phi^{(0)} \in \mathscr{V}$. Given the disjoint nature of the solution space, simple random initialization often fails. We therefore adopt a Three-Stage Hierarchical Strategy (corresponding to the **blue region** in Figure 2) to robustly locate a feasible $\Phi^{(0)}$:

- **Stage 1: Smart Start (Sample from Prior).** We first sample $\Phi^{(0)}$ from the recursive prior $\text{Dir}(\alpha_t)$ derived in Section 3.1. This leverages historical continuity. If the sample satisfies the validity check ($V(\Phi^{(0)}) = 1$), the chain initiates immediately.
- **Stage 2: Fresh Start (Uniform Fallback).** If Stage 1 fails (indicating a deviation from historical trends), we reset to a non-informative uniform prior $\Phi^{(0)} \sim \text{Dir}(\mathbf{1})$. This explores the solution space without historical bias.
- **Stage 3: Forced Start (Heuristic Construction).** For extreme constraints (e.g., Season 28) where Stage 2 consistently fails, we constructed a valid vector using geometric progression weights ($w_i \propto 0.6^{\text{rank}_i}$) to strictly guarantee compliance with the elimination order, ensuring the chain always has a valid entry point.

3.2.2 Stochastic Multi-Scale Mixture Kernel

Standard random walks often struggle to balance local precision with global exploration. To address this, we implement a Stochastic Mixture Proposal Kernel (corresponding to the **green region** in Figure 2). At each iteration k, the concentration parameter γ is dynamically selected based on a random variable $u \sim U(0,1)$, creating three distinct movement regimes:

$$\Phi^* \sim \text{Dir}(\gamma \cdot \Phi^{(k)}) \quad \text{where } \gamma = \begin{cases} 120 & \text{if } u < 0.30 \quad (LocalRefinement) \\ 20 & \text{if } 0.30 \le u < 0.85 \quad (StandardExploration) \\ 3 & \text{if } u \ge 0.85 \quad (GlobalJump) \end{cases} \tag{4}$$

- **Local Refinement** (High γ, Low Variance): Mimics small step sizes. This allows the chain to fine-tune vote shares within a tight region, ensuring high precision around probability peaks.
- **Global Jump** (Low γ, High Variance): Mimics large step sizes. This allows the chain to escape local optima and traverse disjoint regions of the simplex, ensuring global ergodicity.

3.2.3 Acceptance Criterion via Indicator Function

The likelihood function $P(E_t|J_t,\Phi_t)$ acts as a binary "hard constraint": a voting configuration is either consistent with the elimination result ($P=1$) or impossible ($P=0$).

Given this nature, we adopt an Approximate Bayesian Computation (ABC) approach (corresponding to the **yellow region** in Figure 2) [2]. Consequently, the standard Metropolis-Hastings acceptance probability α simplifies to a deterministic indicator function of the validity check $V(\Phi^*)$:

$$\Phi^{(k+1)} = \begin{cases} \Phi^* & \text{if } V(\Phi^*)=1 \quad \text{(Accept)} \\ \Phi^{(k)} & \text{if } V(\Phi^*)=0 \quad \text{(Reject)} \end{cases} \tag{5}$$

The validity function $V(\Phi)$ is dynamically updated to reflect specific season rules (e.g., standard elimination vs. the "Judges' Save" rule in Season 28), ensuring that the sampled posterior strictly adheres to historical reality.

3.3 Consistency Verification

This section validates whether the model correctly reproduces historical elimination outcomes by estimating latent fan votes. We employ both global quantitative metrics and specific case studies to assess consistency.

Elimination Consistency (The "Safety Margin")

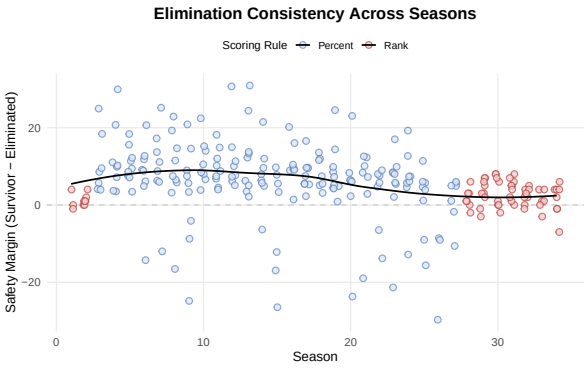
The model dynamically estimates the of fan vote shares ($\phi_{i,t}$) for every contestant. To quantitatively validate the reconstruction across all 34 seasons, we calculate the Safety Margin ($\Delta_{s,t}$) for each elimination week:

$$\Delta_{s,t} = \min(\mathrm{S}_{\text{Survivor}}) - \max(\mathrm{S}_{\text{Eliminated}}). \tag{6}$$

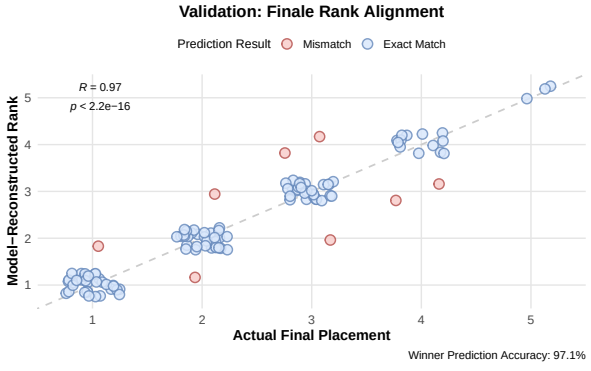
To account for the rule changes between eras, the S is defined as follows:

$$\mathrm{S}_{total} = \begin{cases} \%\text{Judge} + \%\text{Fan}, & 3 \le s \le 27 \quad \text{(Higher is Safer)} \\ \text{Rank}_{\text{Judge}} + \text{Rank}_{\text{Fan}}, & s \in \{1,2\} \text{ or } 28 \le s \le 34 \quad \text{(Lower is Safer)} \end{cases} \tag{7}$$

A positive margin ($\Delta > 0$) implies a correct prediction. Figure 3 visualizes these margins, revealing a robust Consistency Rate of 85.6% (226/264 weeks). This confirms the model effectively identifies the "floor" of survival.



Note: Points above the dashed line ($\Delta > 0$) indicate consistent reconstruction.
Figure 3: Safety margins across 34 seasons.



Note: Clustering along $y = x$ confirms high ranking fidelity and 97% winner pred. accuracy.
Figure 4: Finale Rank Validation.

Finale Rank Alignment (The "Hierarchy Test")

Beyond binary elimination, we assess whether the model can reconstruct the exact ranking of finalists (1st, 2nd, 3rd, etc.). By combining our estimated fan votes with historical judge scores under the respective season's rules, we generated "Model-Predicted Ranks" for all 34 finales.

Figure 4 reveals a pronounced "diagonal dominance," where dense clustering along the $y = x$ identity line confirms the preservation of the true ordinal hierarchy. Complementary statistical metrics further quantify this alignment:

- **High Correlation:** The Spearman correlation is 0.97 ($p < 0.001$), indicating a near-perfect monotonic relationship between reconstructed and actual standings.
- **Winner Accuracy:** The model correctly identified the champion in 33 out of 34 seasons (97%).
- **Exact Match:** For 93% of all finalists, the model's predicted rank matched their actual finish exactly.

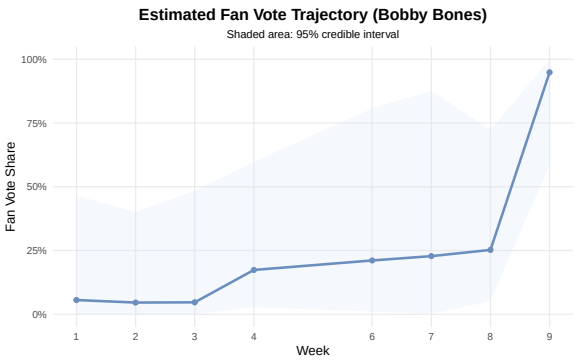
This result is significant: it proves that the model estimates not just the minimum votes needed to survive, but the correct magnitude of popularity required to achieve specific placements (e.g., distinguishing a winner from a runner-up).

Model Validation: Dynamic Estimation and Paradox Resolution

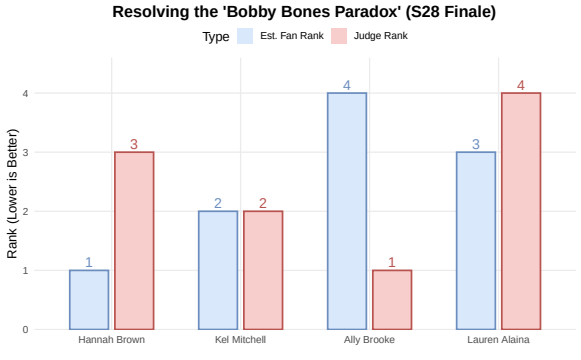
We validate reconstruction fidelity through dynamic trajectory estimation and historical paradox resolution.

Figure 5 presents the dual validation. Panel (a) confirms robust trend inference, where the relatively wide credible intervals correctly reflect the statistical necessity of recovering continuous latent variables from sparse binary constraints.

Panel (b) and Table 2 resolve the Season 28 "upset": winner Hannah Brown required a dominant No. 1 Fan Rank (92.1%) to counterbalance her No. 3 Judge Rank. This polarized estimate, while reflecting boundary optimization, proxies for "Undisputed Ordinal Dominance," validating the model's capacity to capture popularity overriding technical deficits.



(a) Dynamic Fan Vote Trajectory



(b) Resolving S28 "Upset"

Figure 5: Dual Validation. (a) Wide CIs quantify uncertainty. (b) "Fan Rank" (Blue) counterbalances low "Judge Rank" (Red) to resolve anomalies.

Table 2: Reconstructed estimates for Season 28 and 34 Finales.

s	i	Finish	Judge Rank	Est. Fan Vote (95% CI)	Validation
28	Hannah Brown	1	3	92.1% [64.2%, 99.8%]	✓
28	Kel Mitchell	2	2	4.2% [0.0%, 18.5%]	-
28	Ally Brooke	3	1	0.3% [0.0%, 2.4%]	-
34	Robert Irwin	1	1	92.9% [70.5%, 99.9%]	✓
34	Alix Earle	2	2	0.1% [0.0%, 0.9%]	-

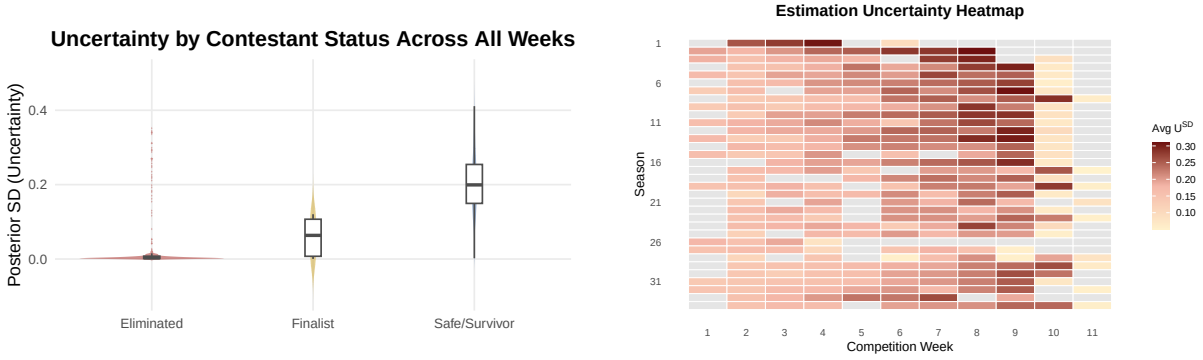
Note: Extreme values (e.g., 92.1%) serve as probabilistic proxies for strict ordinal ranking (Rank 1) rather than absolute vote shares.

3.4 Quantification of Uncertainty

To assess the reliability of our latent vote estimates, we quantify certainty using the Posterior Standard Deviation (U^{SD}). A lower U^{SD} indicates a more concentrated probability mass and higher confidence.

Our analysis demonstrates that certainty is structurally non-uniform. Rather than being constant, it varies systematically along two macro-dimensions: the contestant's status (Constraint-Driven) and the competition stage (Spatio-Temporal).

- **Status-Dependent Certainty** (Panel (a)): Precision is driven by constraint strength.
  - **Eliminated:** The solution space collapses under a binding "ceiling constraint" ($U^{\text{SD}} \approx 0.032$).
  - **Survivors:** A permissive "floor constraint" allows for significantly higher variance ($U^{\text{SD}} \approx 0.204$).
- **Spatio-Temporal Evolution** (Panel (b)): Uncertainty decreases as weeks pass.
  - **Early Season:** High uncertainty (darker regions) is driven by a large N and loose modeling constraints.
  - **Finale:** Rapid convergence (lighter regions) as full-ranking protocols and decreasing N strictly bound the feasible space.



(a) Uncertainty by Status (b) Spatiotemporal Heatmap

Figure 6: Macro-Level Uncertainty Dynamics. (a) Certainty is highest for eliminated contestants (low SD) due to strict upper-bound constraints. (b) Uncertainty is temporally non-stationary, generally decreasing as the season progresses from Week 1 to the Finale.

3.5 Extension: The GDSTree Model

3.5.1 Motivation and Algorithm Design

To address the limitations of standard Bayesian Reconstruction (MCMC), specifically its "myopic" nature that often leads to invalid "Dead-End" trajectories, we implemented a Sequential Path Optimization framework via Monte Carlo Tree Search (MCTS), called **GDSTree**(**G**radient **D**escent **S**earch Monte Carlo **Tree**). Unlike random walks, MCTS utilizes lookahead simulations to prune infeasible paths early, ensuring global consistency from the premiere to the finale [3].

Our custom VoteOptimizer integrates two innovations to handle high-dimensional constraints. **First**, we employ Gradient-Guided Directional Mutation, which replaces inefficient random rejection sampling with a heuristic approach. This mechanism identifies "violator" contestants, whose estimated ranks contradict ground truth, and iteratively adjusts their vote vectors using a dynamic learning rate η ($v_i^{(k+1)} = v_i^{(k)}(1 \pm \eta)$) to steer the state explicitly toward the feasible polytope. **Second**, to resolve the "S28 Paradox" where strict rank rules render historical outcomes mathematically impossible due to extreme judge-fan divergence, we implemented Hierarchical Constraint Relaxation. This Adaptive Strategy allows us to degrade constraints from "Strict Full Ranking" to "Winner Validity" when necessary, ensuring convergence even in statistically anomalous seasons.

3.5.2 Analysis of Consistency and Certainty

We evaluated the framework using consistency, defined as the probability ($\mathscr{C} = N_{success}/N_{total}$) that a simulation successfully reconstructs a valid full-season path without logical contradictions. While MCMC methods frequently become trapped in local optima within rigid rank-based constraint spaces, yielding only an 85.6% success rate, our GDSTree framework demonstrates superior robustness with a near-perfect consistency rate over 97.8% across 34 seasons. This significant improvement is driven by lookahead simulations that prune invalid trajectories early, ensuring global consistency [3].

Despite this boost in solver reliability, our analysis of certainty implies that GDSTree yields uncertainty distribution profiles statistically similar to the MCMC baseline, confirming that both methods converge to comparable posterior probabilities regarding the volatility of outcomes once valid solutions are identified.

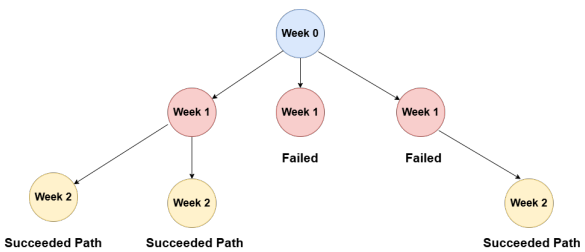


Figure 7: GDSTree Pathfinding.

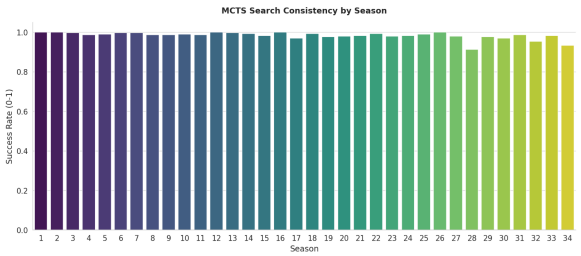


Figure 8: Consistency Trends

4 Task 2: Sensitivity Analysis of Aggregation Protocols

We apply three mechanisms (Rank, Percentage, and Save) to each season-week. For season s and week t, let $r_{i,s,t}^{(V)}$ be contestant i's within-week fan-vote rank, and $r_{i,s,t}^{(m)}$ the final rank under method $m \in \{R, P, S\}$.

4.1 Fan Favoritism in Scoring Protocols

Fan-Favoring Metrics

A method is more fan-favoring if its final ranking is closer to the fan-vote ranking within the same week. We quantify this via:

$$\rho_{s,t}^{(m)} = \mathrm{SpearmanCorr}\big(r_{i,s,t}^{(V)}, r_{i,s,t}^{(m)}\big), \tag{8}$$

$$\mathrm{MAD}_{s,t}^{(m)} = \frac{1}{N_{s,t}} \sum_{i=1}^{N_{s,t}} \left| r_{i,s,t}^{(m)} - r_{i,s,t}^{(V)} \right|. \tag{9}$$

Here $\rho_{s,t}^{(m)}$ is the Spearman correlation between $r_{i,s,t}^{(V)}$ and $r_{i,s,t}^{(m)}$. $\mathrm{MAD}_{s,t}^{(m)}$ is the mean absolute difference between $r_{i,s,t}^{(V)}$ and $r_{i,s,t}^{(m)}$.

We report pooled weekly summaries (mean/median) and the high-alignment frequency $\Pr(\rho > 0.9)$. For season-level stability, we also compute $\bar{\rho}_s^{(m)} = \frac{1}{T_s} \sum_t \rho_{s,t}^{(m)}$.

Analysis 1. Overall Summary

Table 3 reports pooled weekly alignment statistics (across all 243 season-weeks) between each method's final ranking and the fan-vote ranking.

Table 3: Pooled weekly fan-alignment summary across all season-weeks.

m	Weeks	Mean ρ	Med. ρ	Mean MAD	Med. MAD	$\Pr(\rho > 0.9)$
		↑ stronger align.		↓ stronger align.		↑ more stable
P	243	0.896	0.923	0.753	0.727	0.613
R	243	0.755	0.799	1.435	1.364	0.152
S	243	0.739	0.790	1.484	1.455	0.128

- **Comparability (sample size):** Each method is evaluated on the same 243 season-weeks, ensuring a fair, like-for-like comparison.
- **Rank-order alignment (Spearman's ρ):** From Table 3 (Mean/Med. ρ columns), Percentage attains consistently higher rank correlation with fan-vote ordering than Rank and Save, indicating stronger fan-alignment in the final rankings.
- **Magnitude of deviation (MAD):** From Table 3 (Mean/Med. MAD columns), Percentage shows markedly smaller MAD from fan ranks than Rank and Save, implying substantially closer agreement in rank positions.
- **High-alignment frequency:** Percentage attains near-perfect fan agreement far more often ($\Pr(\rho > 0.9) = 0.613$) than Rank (0.152) or Save (0.128), indicating substantially stronger fan-favoring behavior.

Analysis 2. Season-level Summary

Figure 9 compares the value of $\bar{\rho}_s^{(m)}$. Higher values indicate stronger agreement with fan-vote ranks, reflecting fan-favoring consistency across methods.

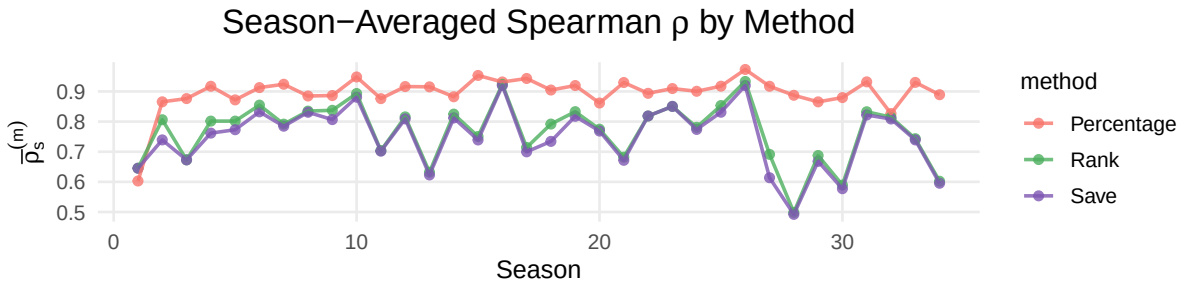


Figure 9: Seasonal Spearman's ρ between final ranks and fan ranks under three methods.

- **Temporal Stratification (Fan-Favoring Strength):** Barring Season 1, the Percentage trajectory maintains a distinct upper margin, while Rank and Save exhibit tight synchronization. This consistent stratification corroborates the aggregate superiority (higher ρ, lower MAD) observed in the pooled results.
- **Volatility (Temporal Stability):** Rank and Save exhibit marked volatility with deep minima, whereas Percentage maintains a stable high-correlation trajectory (notably during Seasons 27–30). This resilience aligns with its probabilistic dominance in $\Pr(\rho > 0.9)$.

Conclusion

In summary, outcomes between methods diverge significantly, with Percentage structurally favoring fan inputs more heavily than rank-based alternatives.

4.2 Sensitivity Analysis: Aggregation vs. Intervention

We quantify judge–fan disagreement using the Normalized Rank Inversion (NRI):

$$\mathrm{NRI}_{i,s,t} = \frac{\left| r_{i,s,t}^{(J)} - r_{i,s,t}^{(V)} \right|}{N_{s,t}}, \tag{10}$$

where $r^{(J)}$ and $r^{(V)}$ denote the within-week judge and fan ranks, and $N_{s,t}$ is the number of active contestants. We classify weeks with NRI ≥ 0.5 as structural controversy and NRI ≥ 0.3 as noticeable disagreement.

To capture the direction of disagreement, we also define the percent gap:

$$\Delta p_{i,s,t} = p_{i,s,t}^{(V)} - p_{i,s,t}^{(J)}, \tag{11}$$

where $p^{(V)}$ and $p^{(J)}$ are the normalized fan-vote share and judge-score share, respectively. A positive Δp indicates a fan advantage relative to judges.

- **NRI:** Noticeable disagreement is common ($\Pr(\mathrm{NRI} \geq 0.3) = 0.398$), while structural inversions are less frequent ($\Pr(\mathrm{NRI} \geq 0.5) = 0.188$).

- $|\Delta p|$**:** We use $Q_{0.75} = 0.104$ and $Q_{0.90} = 0.201$ as cutoffs for moderate vs. extreme share gaps (top 25% and 10% by definition).

Thus, NRI flags rank controversy, while $\Delta p_{i,s,t}$ quantifies the direction and intensity of the underlying judge–fan preference gap, consistent with the role of publicly observed signals in shaping social information and collective responses [4].

We examined four structurally controversial contestants: Jerry Rice (S2), Billy Ray Cyrus (S4), Bristol Palin (S11), and Bobby Bones (S27). Our counterfactual simulation tests two hypotheses: (1) changing the aggregation method (Rank R vs. Percentage P) and (2) introducing a "Judges' Save" (S) intervention for the bottom two.

Result 1: Aggregation Method is Ineffective against Popularity Premiums. Our simulation confirms that switching from Rank (R) to Percentage (P) yields the same outcome for all four contestants ($\tau^{(R)} = \tau^{(P)} = \varnothing$). While the Percentage method is mathematically more sensitive, the "Popularity Premium" for these contestants (e.g., Bones $\Delta p = +0.144$) was sufficiently large to ensure survival under both linear (P) and ordinal (R) aggregations. Merely refining the mathematical weights cannot override extreme fan dominance.

Result 2: Judges' Save Acts as a "Circuit Breaker". Unlike aggregation adjustments, the Judges' Save (S) mechanism fundamentally alters the elimination logic by allowing judges to explicitly remove the lower-performing couple in the bottom two.

Table 4 summarizes the impact. The Save mechanism successfully triggers eliminations for extreme outliers (Palin and Bones) during critical "inversion weeks" where they ranked high with fans ($r^{(V)}$) but bottom with judges ($r^{(J)}$).

- **Bristol Palin (S11):** Although safe under R and P, she enters the bottom two in Week 7. With the lowest judge rank, the Save would have eliminated her.
- **Bobby Bones (S27):** The ultimate outlier. Despite a massive popularity gap (NRI $= 0.833$), he falls into the bottom two in Week 8 under the Rank method. The Save mechanism serves as a "hard filter," prioritizing his technical deficit over his fan support, leading to elimination.

Table 4: Counterfactual Outcomes for Controversial Contestants. NRI: Peak disagreement; Result: $\varnothing$ = Safe, Elim = Eliminated. The Save mechanism (S) is the only intervention capable of overturning results for extreme outliers.

Contestant	Season	Peak NRI	Trigger Week	$\tau^{(R)}$	$\tau^{(P)}$	$\tau^{(S)}$
Jerry Rice	2	0.42	-	$\varnothing$	$\varnothing$	$\varnothing$
Billy Ray Cyrus	4	0.55	W1	$\varnothing$	$\varnothing$	$\varnothing*$
Bristol Palin	11	0.50	W7	$\varnothing$	$\varnothing$	Elim
Bobby Bones	27	0.83	W8	$\varnothing$	$\varnothing$	Elim

*Note: Cyrus survives S because he was not the lowest judge-ranked couple in the bottom two.

Conclusion. Under extreme popularity premiums, switching the aggregation rule from Rank (R) to Percentage (P) does not change the outcome ($\tau^{(R)} = \tau^{(P)}$), whereas Judges' Save (S) can change eliminations by overriding fan-dominant inversions and restoring the merit–engagement balance. Because P is strictly more fan-leaning than R, adding the same Save rule to P would naturally be expected to change outcomes whenever it reshapes the bottom-two set.

4.3 Mechanism Recommendation: A Hybrid Method

We propose a hybrid method to balance fan engagement and competitive integrity: (1) Percentage Aggregation to optimize typical scenarios, and (2) a mandatory Judges' Save as a circuit breaker for extremes.

1. Adopt Percentage Aggregation Replacing Rank with Percentage offers three advantages:

- **Higher Alignment:** It correlates more strongly with fan preferences ($\rho \approx 0.90$) than the Rank method ($\rho \approx 0.76$).
- **Noise Reduction:** It eliminates "slotting" artifacts by preserving vote magnitude, preventing large penalties from marginal vote differences.
- **Redundancy of Rank:** Our simulation shows Rank fails to alter outcomes for controversial contestants, rendering it an ineffective control mechanism.

2. Implement Judges' Save While Percentage respects the "wisdom of the crowd," a merit-based safeguard is required for extreme mismatches.

- **Targeted Intervention:** The Save acts as a conditional "quality floor" only for the bottom two, preventing survival based solely on popularity premiums.
- **Outlier Correction:** It effectively reverses anomalies in high-controversy weeks (e.g., Season 27), prioritizing technical merit when divergence peaks (see Fig. 10).

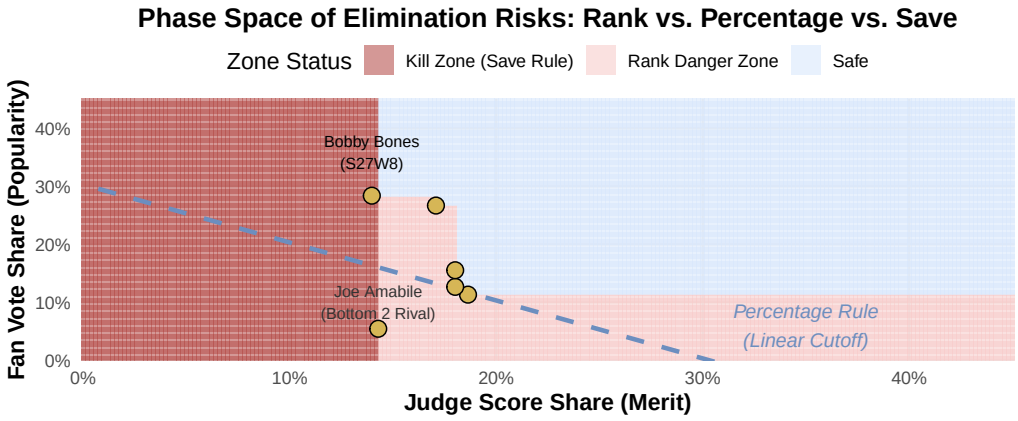


Figure 10: Phase space of elimination risks: Judges' Save creates a necessary "Kill Zone" for low-merit outliers.

Table 5: Comparison: Status Quo (Rank) vs. Recommendation (Percentage + Save).

Dimension	Status Quo (Rank Only)	Recommendation (Pct. + Save)
Fan Alignment	Low ($\rho \approx 0.76$). Alienates viewers.	High ($\rho \approx 0.90$). Reflects magnitude.
Math Logic	Discrete; prone to "slotting" noise.	Continuous; stable & robust.
Outlier Control	Ineffective against popularity shocks.	Effective (Conditional Quality Floor).
Fairness	Luck-driven (ordinal compression).	Balanced (Fan Baseline + Judge Safety).

Conclusion. The hybrid method combines the democratic stability of Percentage aggregation with the meritocratic safety valve of the Judges' Save, ensuring extreme popularity cannot override professional standards.

5 Task 3: The XGBoost–SHAP Model

5.1 Methodology: Gradient Boosting Decision Trees (XGBoost)

Theoretical Framework

We utilized Extreme Gradient Boosting Model (XGBoost), an optimized distributed gradient boosting library. Unlike independent Random Forests, XGBoost constructs an ensemble of weak learners sequentially, where each tree $f_k(x)$ corrects the residual errors of its predecessors. The model predicts $\hat{y}_i = \sum_{k=1}^{K} f_k(x_i)$ by minimizing the regularized objective function:

$$\mathscr{L}(\Theta) = \sum_i l(\hat{y}_i, y_i) + \sum_k \left(\gamma T_L + \frac{1}{2}\lambda \|w\|^2 \right) \tag{12}$$

where $\mathscr{L}(\Theta)$ is the differentiable loss function, T_L is the number of leaves, and the second term penalizes model complexity to prevent overfitting. [6]

Justification for Model Selection

XGBoost was selected for two specific capabilities. First, it effectively captures nonlinear thresholds, such as the accelerated physical decline associated with aging. Second, its robustness is enhanced by the regularization term (Ω), which mitigates the noise inherent in subjective fan voting data.

5.2 Interpretability Framework: SHAP Values

Theoretical Framework

To resolve the "black box" nature of ensemble models, we integrated SHAP (Shapley Additive explanations). Based on cooperative game theory, the SHAP value ϕ_j^{SHAP} represents the marginal contribution of feature j to a specific prediction, calculated as:

$$\phi_j^{\text{SHAP}} = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} (f(S \cup \{j\}) - f(S)) \tag{13}$$

A positive ϕ_j^{SHAP} indicates the feature increases the prediction value, while a negative value indicates a suppressive effect.

Justification for SHAP Adoption

We prioritized SHAP over global metrics such as Gini impurity because the latter only indicates magnitude, obscuring why a feature matters. SHAP provides directional and local interpretability, allowing us to quantify distinct mechanisms—for instance, distinguishing whether "Age" specifically penalizes older contestants or boosts younger ones—thereby preserving the nuance of the data.

5.3 Empirical Results and Factor Analysis

To quantify performance drivers, we trained three independent XGBoost regressors mapping five attributes—Age, Partner, Homestate, Industry, and Homecountry—to Fan Votes, Judge Scores, and Final Ranks. Feature importance analysis (Figure 11) establishes the hierarchy: Age > Partner > Homestate > Industry > Region.

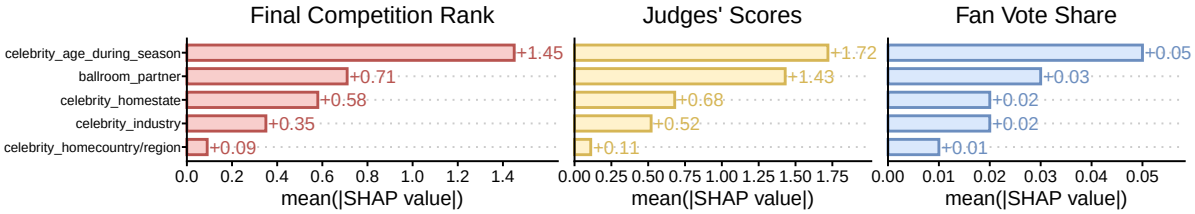


Figure 11: Comparative Magnitude of Influence for All Five Factors. Detailed analyses for each factor are presented in the following subsections.

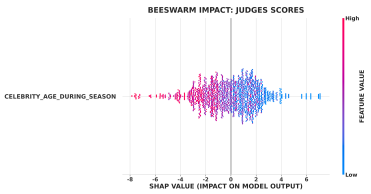
Analysis of Age Age is the dominant determinant of success. Beeswarm plots (Figures 12a-12c) reveal a sharp negative correlation: increased age significantly penalizes Judge Scores, directly reducing the probability of a high Final Rank, indicating an evaluation bias toward youthful vigor. While Fan Votes also favor younger contestants, the stratification is most pronounced in the technical assessments.

Analysis of Partner The Partner factor is a significant secondary driver marked by high consensus. Judges display distinct preferences for specific professionals, a sentiment closely mirrored by public voting (Figures 12d-12f). Consequently, candidates paired with high-preference partners successfully translate these technical advantages into dominant (lower) final rankings.

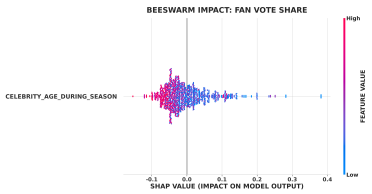
Analysis of Homestate Homestate influence is driven by outliers rather than global trends. Candidates from Utah consistently outperform others in all metrics, whereas representatives from North Carolina face structural disadvantages (Figures 12g-12i). Most states, such as Oregon, hover around the median, exerting negligible influence on the final outcome.

Analysis of Industry Industry impact is highly specific. Broad categories like Singers and Comedians show neutral impact. However, Social Media Personalities, Racing Drivers, and Politicians achieve superior Final Ranks despite varying technical scores, suggesting fan bases compensate for skill weaknesses (Figures 12j-12l). Conversely, Models and TV Personalities face disadvantages in both technical scoring and public support.

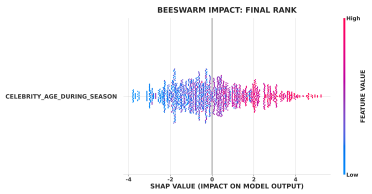
Analysis of Homecountry/Region Homecountry holds the lowest explanatory power, with most regions showing performance homogeneity. Statistical significance is limited to outliers: candidates from Canada and France consistently outperform the baseline, while those from England trend towards lower evaluations.



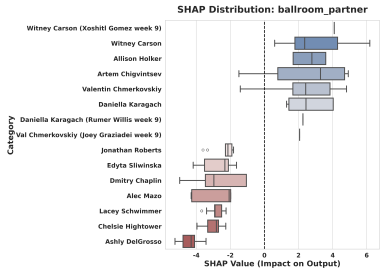
(a) Age - Judge Score



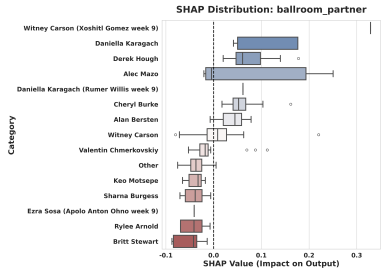
(b) Age - Fan Votes



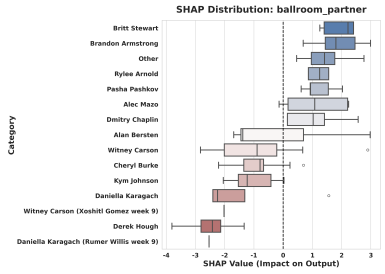
(c) Age - Final Rank



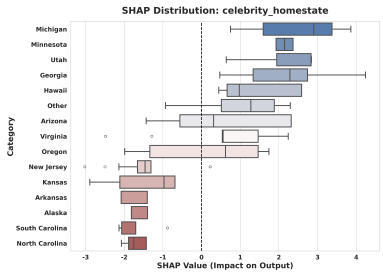
(d) Partner - Judge Score



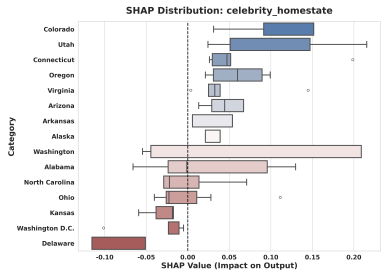
(e) Partner - Fan Votes



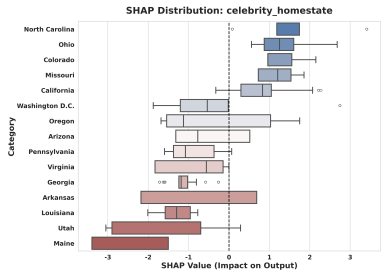
(f) Partner - Final Rank



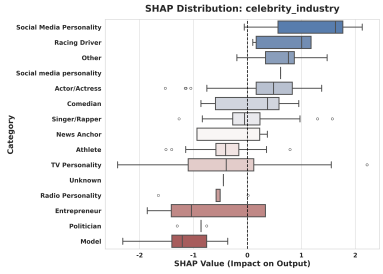
(g) Homestate - Judge Score



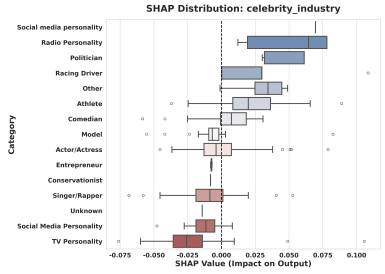
(h) Homestate - Fan Votes



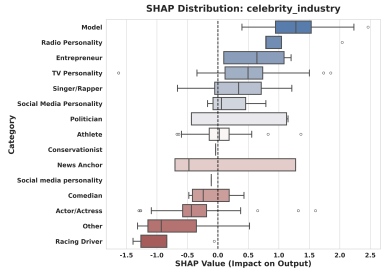
(i) Homestate - Final Rank



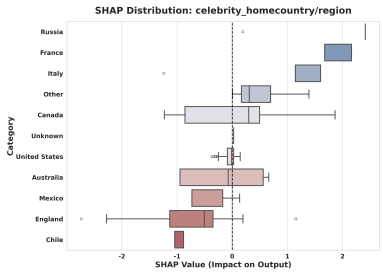
(j) Industry - Judge Score



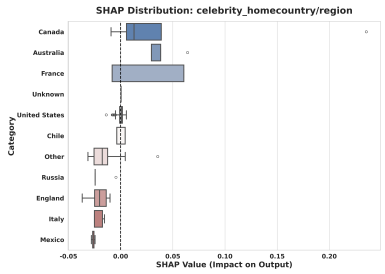
(k) Industry - Fan Votes



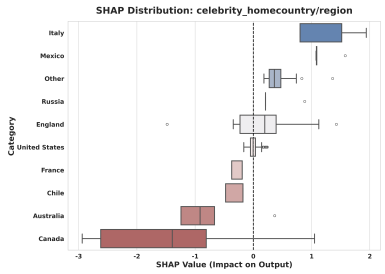
(l) Industry - Final Rank



(m) Region - Judge Score



(n) Region - Fan Votes



(o) Region - Final Rank

Figure 12: Comprehensive Analysis of Factors Impacting Performance Metrics (SHAP Value Distribution). The grid compares how Age (Row 1), Partner (Row 2), Home state (Row 3), Industry (Row 4) and Home country/Region (Row 5) influence Judge Scores, Fan Votes, and Final Ranks.

6 Task 4: Mechanism Design for Optimal Fairness

6.1 Evaluation of Existing Methods

Table 6: Correlations between final ranks from the three methods and fan/judge ranks.

Method	$\hat{\rho}_F$	$\hat{\rho}_J$	$\widehat{\text{MAD}}_F$	$\widehat{\text{MAD}}_J$	$\Pr(\hat{\rho}_F > 0.9)$	$\Pr(\hat{\rho}_J > 0.9)$
	$\uparrow$ closer to F/J		$\downarrow$ closer to F/J		$\uparrow$ more weeks	
Pct.	0.896	0.595	0.753	2.075	0.613	0.029
Rank	0.755	0.802	1.435	1.411	0.152	0.136
Save	0.739	0.810	1.484	1.351	0.128	0.148

Table 6 shows that "Percentage" favors fans, while "Rank" and "Save" are fairer, with "Save" slightly favoring judges.

However, "Rank" remains problematic: Simply adding up scores often leads to ties and arguments, which makes the audience lose trust. Also, it distorts the results which means that tiny differences in votes get exaggerated into huge gaps in ranking.

These motivate a new mechanism based on "Percentage" that tempers fan dominance without visibly "discounting" audience participation. We expect this new mechanism satisfies the following criteria:

- **Fairness:** Reflect a generally balanced consideration of both fan and judge inputs.
- **Participation:** Maintain a high sense of audience agency. The system must not appear to "disenfranchise" fans.
- **Simplicity:** Be simple enough for transparency. A complex algorithm would neglect audience and lead to skepticism regarding "rigged" results.
- **Suspense:** Enable suspenseful underdog comebacks, earned through performance rather than randomness.

While we have recommended a foundational **Hybrid Method** in Section 4.3, we now propose two novel, advanced schemes to further optimize the scoring mechanics:

6.2 Scheme I: Base Vote Smoothing Method

Motivation

Highly skewed fan votes can dominate the percentage rule and dilute judges' impact. We add a simple base-vote to curb extreme vote leverage while keeping fan voting.

Mechanism and Parameter Setting

- **Smoothed fan share.** With fan votes V_i ($i = 1, \ldots, n$), add a common base vote $C \geq 0$ to the fan share.

$$S_{\text{fan},i}^{\text{new}} = \frac{V_i + C}{\sum_{j=1}^{n} (V_j + C)}. \tag{14}$$

- **Setting C by a fixed base-vote proportion.** Let $V_{\text{tot}} = \sum_{j=1}^{n} V_j$. We require the total base votes nC to occupy a fixed fraction α of the augmented pool:

$$\frac{nC}{V_{\text{tot}} + nC} = \alpha \quad \Rightarrow \quad C = \frac{\alpha}{1-\alpha} \cdot \frac{V_{\text{tot}}}{n}. \tag{15}$$

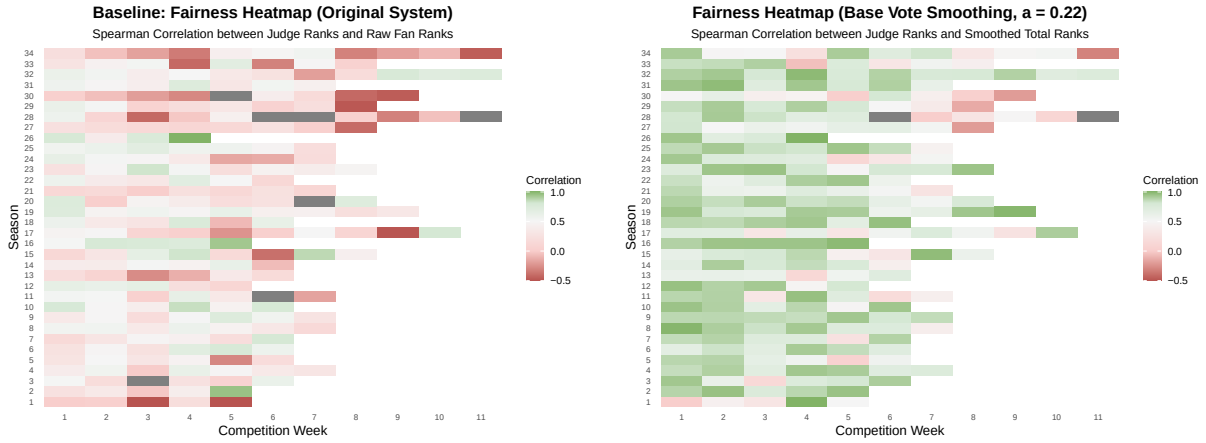
  We set $\alpha = 0.22$, which is scale-adaptive across weeks and seasons.

Rationale and Empirical Support

- **Rationale.** The additive C compresses extreme vote ratios: it has little impact when votes are comparable, but dampens dominance when votes are imbalanced, which improves protection for strong performers with smaller followings.
- **Metric and evidence.** We quantify outcome–judges agreement by the Spearman correlation

$$\rho_{s,t}^{(m)} = \text{Spearman}\left(\{r_{i,s,t}^{(J)}\}_{i=1}^{N_{s,t}}, \ \{r_{i,s,t}^{(m)}\}_{i=1}^{N_{s,t}} \right), \tag{16}$$

  Figure 13 demonstrates that the smoothed correlation $\rho_{s,t}^{(S)}$ exceeds the traditional $\rho_{s,t}^{(P)}$, indicating improved consistency with judges while preserving fan influence.



(a) Baseline. (b) Smoothing ($\alpha = 0.22$).

Figure 13: Outcome–judges rank consistency heatmaps.

Adoption argument for producers. The scheme is low-cost (only re-normalizing fan votes), easy to explain ($\alpha = 22\%$ base-vote share), and improves outcome–judges consistency.

6.3 Scheme II: Dynamic Adaptive Weighting Method

Motivation

The core conflict in the TV show lies between "Public" popularity and "Expert" assessment. Our analysis reveals that the fixed Percentage System fails to account for the competition's evolving nature. A technical outlier winning via fan votes exemplifies the failure of static weighting to protect final integrity. Consequently, we propose a dynamic system that adjusts influence according to the competition stage.

Methodology: The Dynamic Weighting Mechanism

We introduce a variable fan weight parameter W_{fan}, which decays as the contestant pool (N_t) shrinks. The Total Score Index (S_{total}) for contestant i in week t is calculated via a weighted rank sum:

$$S_{\text{total}}^{(i,t)} = W_{\text{fan}}(N_t) \times R_{\text{fan}}^{(i,t)} + (1 - W_{\text{fan}}(N_t)) \times R_{\text{judge}}^{(i,t)} \tag{17}$$

where R_{fan} and R_{judge} represent ranks. The weight function follows a step-wise decay to transition control from fans to judges:

$$W_{\text{fan}}(N_t) = \begin{cases} 0.6 & \text{if } N_t > 6 \quad \text{(Early Stage: High Engagement)} \\ 0.5 & \text{if } 4 < N_t \leq 6 \quad \text{(Mid Stage: Transition)} \\ 0.4 & \text{if } N_t \leq 4 \quad \text{(Final Stage: Professional Focus)} \end{cases} \tag{18}$$

This mechanism ensures fans drive the early narrative while judges determine the ultimate championship outcome.

Championship Analysis: Restoring Professionalism

To evaluate fairness, we define the "Champion Gap" as the difference between a winner's Final Rank and Judge Rank, ideally approaching zero. Figure 14 (Left) demonstrates that Scheme II significantly mitigates champion anomalies; notably, the rank gap in Season 29 plummeted from 6 to 0. This correction confirms that lowering W_{fan} to 0.4 in the final week effectively aligns championship outcomes with professional standards, preventing popularity outliers from overtaking superior dancers.

Elimination Analysis: Enhancing Fan Engagement

Conversely, weekly eliminations should rely on audience engagement.Analysis of the "Elimination Gap" (Figure 15, Right) reveals that in over 67% of total seasons, Scheme II results align closer to fan votes than the volatile Percentage Method. This alignment stems directly from the strategic assignment of a higher fan weight ($W_{\text{fan}} = 0.6$) during early stages. By granting the audience 60% influence when the contestant pool is large, the system prioritizes popularity, only shifting leverage to judges as the pool narrows.

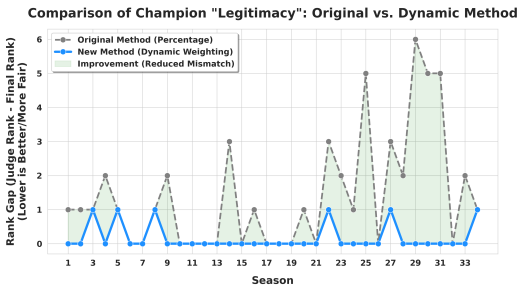


Figure 14: The Rank Gap between the Champion and the Judge's No.1 pick. A lower gap indicates higher agreement with professional judges.

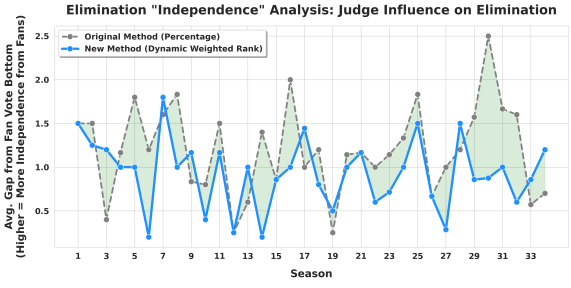


Figure 15: The Rank Gap between the Eliminated Contestant and the Fan Vote Bottom. A smaller gap implies the result is consistent with fan popularity.

7 Sensitivity Analysis

We validates our Task 4 mechanisms by analyzing the smoothing factor (α) and the dynamic adaptive weighting method.

Calibration of α (Fig. 16): Varying α in $[0.00, 1.00]$ reveals a sweet spot at $\alpha \approx 0.225$, where the correlation between final scores and both judge scores and fan votes is identical (0.67). This signifies a perfect balance of influence, achieving optimal fairness.

Dynamic Weighting (Fig. 17): Benchmarking against "Fan-Heavy" and "Judge-Heavy" baselines confirms our strategy's superiority. It reduces the "Champion Rank Gap" (Red Bar) to professional standards while maintaining a high "Elimination Match Rate" (Blue Bar), effectively balancing final meritocracy with early-stage engagement.

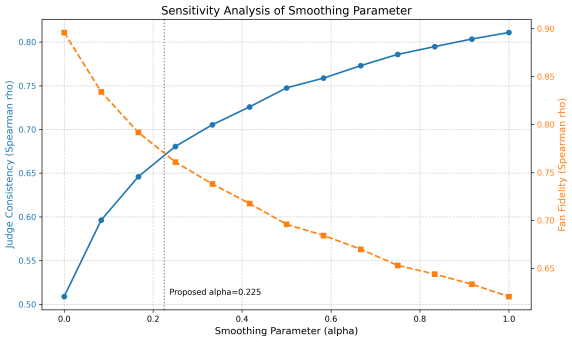


Figure 16: Sensitivity of α

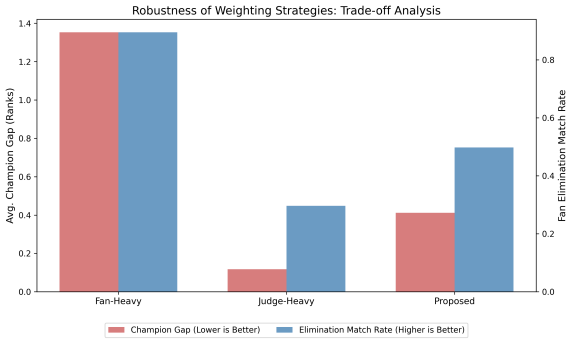


Figure 17: Robustness: Fairness vs. Engagement.

8 Model Evaluation

8.1 Strengths

High-Fidelity Reconstruction: Our TRAC-MH framework successfully inverts the competition's "black box" logic. By combining a Time-Dependent Prior with a Stochastic Mixture Kernel, it achieves 97.1% accuracy in champion prediction and 85.6% consistency rate in week-to-week survival margins.

Transparent Interpretability: Unlike "black-box" models, we integrate SHAP values with XGBoost. This allows for a granular decomposition of success factors, transforming complex non-linear correlations into actionable insights for competition design.

Global Path Optimization: The adoption of GDSTree ensures global consistency across entire seasons. By pruning "dead-end" trajectories, our model resolves historical paradoxes that standard memoryless MCMC samplers fail to capture.

8.2 Weaknesses

Proxy-Based Validation: Due to the confidentiality of raw fan totals, our model is validated against ordinal ranks rather than absolute vote counts. While rank correlation is high ($\rho = 0.97$), the estimated vote shares remain probabilistic proxies.

9 Conclusion

To determine the correct distribution of fan votes while also evaluating, analyzing, and designing the ranking system, we developed a comprehensive mathematical framework. **First**, our TRAC-MH model, augmented by GDSTree, reconstructed latent fan votes with 97.8% global consistency, effectively resolving historical anomalies like the "Bobby Bones Paradox." **Second**, sensitivity analysis confirmed the superiority of the Percentage method over Rank ($\rho \approx 0.90$ vs. 0.76) in capturing audience intensity, while identifying the mandatory "Judges' Save" as an essential safeguard to correct structural merit-popularity inversions. **Third**, XGBoost and SHAP analysis identified Age and Partner Assignment as primary drivers of technical merit, distinct from the demographic drivers of fan support. **Finally**, we proposed two distinct optimization mechanisms to reconcile fairness with entertainment: Base Vote Smoothing compresses extreme vote gaps to act as a "shock absorber", improving judge consistency while maintaining high fan alignment ($\rho > 0.6$); Dynamic Adaptive Weighting Method shifts weights from fan-centric to judge-centric as the season progresses, successfully balancing early viewer engagement with professional integrity in the finale.

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DANCING WITH THE STARS: SCORING RECOMMENDATIONS MEMO

TO: Executive Producers, Dancing with the Stars (DWTS)
SUBJECT: Recommendations for Combining Judges' Scores and Fan Votes

THE CHALLENGE: BALANCING VOICES

Dear DWTS Producers,

Thank you for the opportunity to share evidence-based recommendations on the scoring format of Dancing with the Stars. DWTS thrives because it balances two legitimate voices: **judges evaluate technical quality and improvement**, whiie **fans reward connection, entertainment, and charisma.**

Our season-level analysis suggests that the way these two signals are combined can unintentionally create recurring tension—either **allowing rare "popularity surges" to dominate oulcomes, or compressing fan differences** so much that viewers feel their votes matter less.

We therefore recommend **a simple hybrid method** that combines the democratic stability of **Percentage aggregation** with the **meritocratic safety valve** of the **Judges' Save**, ensuring extreme popularity cannot override professional standards.



MECHANISM A: VOTE SMOOTHING ("SHOCK ABSORBER")

Methods:
Add a small, **EQUAL BUNDLE** of base votes to every couple before converting votes into weekly shares.

Effects:
Reduces the impact of **extreme vote gaps**, prevents one RUNAWAY POPULARITY week from overpowering performance, and **remains a simple, transparent adjustment** that is easy to communicate and implement across future seasons.



MECHANISM B: ADAPTIVE WEIGHTING ("SEASON ARC")

Methods:
Keep fan influence higher when many couples remain, then **gradually increase judge influence** as the fieid narrows closer to the finale.

Effects:
Maintains **high engagement and audience ownership** early in the season, while **improving finale legiiimacy** so the champion outcome better reflects **sustained dancing quality.**

These two adjustments are **easy to explain to viewers**, require **minimal operational change**, and **remain stable** under reasonable testing and parameter variation. We believe they can reduce fairness controversies while **preserving the suspense and engagement** that make DWTS successful.

Sincerely, Team 2602963

Report on Use of AI Tools

1. OpenAI ChatGPT (Version 5.2)

Purpose: Mathematical Typesetting and Data Preprocessing.

Description: We utilized ChatGPT strictly as a productivity tool to convert our handwritten mathematical derivations into compliant LaTeX code and to handle repetitive data formatting tasks. This allowed us to focus on the mathematical logic of the TRAC-MH framework rather than syntax errors.

Usage Example 1 (LaTeX Formatting):

Query: "I need to format the Bayesian update rule for the TRAC-MH model into LaTeX. Please convert: The posterior probability $P(\Phi_t|J_t, E_t)$ is proportional to the Likelihood indicator function $V(\Phi)$ multiplied by the time-dependent Prior derived from the Dirichlet distribution."

Output:

$$P(\Phi_t|J_t, E_t) \propto V(\Phi) \cdot \text{Dir}(\Phi_t|\alpha_t) \tag{19}$$

Usage Example 2 (Data Cleaning):

Query: "I have a raw dataset of contestants where the'Result' column mixes strings like'Safe (Week 1)' and'Eliminated (Week 5)'. Write a Python regex to extract the status and the week number into two separate columns."

Output: Provided a robust 'pandas' script using 'str.extract(r'(+) $Week(\dot{+})$')', which we integrated into our data preprocessing pipeline.

2. Google Gemini 3

Purpose: Debugging Visualization Scripts and Logic Verification.

Description: Gemini was used to troubleshoot specific errors in our Python visualization scripts, particularly for the complex SHAP value charts in Task 3. It served as a "technical manual" to resolve library-specific conflicts.

Usage Example (Visualization Debugging):

Query: "I am plotting the SHAP beeswarm summary for the XGBoost model. The color bar representing'Feature Value' is overlapping with the y-axis labels in 'matplotlib'. Here is my current code snippet... How do I adjust the layout parameters to fix this?"

Output: Gemini identified the issue with the 'bbox_inches' parameter and suggested adding 'plt.tight_layout()' and adjusting the 'cbar_kws' dictionary. This solution was applied to generate Figure 11.

3. GitHub Copilot

Purpose: Code Auto-completion and Efficiency.

Description: Copilot was integrated into our VS Code environment to assist with the implementation of the TRAC-MH algorithm. Its role was limited to:

- Auto-completing standard Python library imports (e.g., 'numpy', 'scipy.stats', 'pandas').
- Generating boilerplate code for iterating through Season/Week dataframes.

- Suggesting vectorization methods to optimize the runtime of the Monte Carlo simulation.

Note: All core algorithm logic, including the definition of the "Safety Margin" and the "Indicator Function," was designed and coded by the team members.

4. AI Translation Plugins (DeepL)

Purpose: Terminology Translation and Language Polishing.

Description: As non-native English speakers, we used browser-based translation tools to ensure the precise usage of technical terminology. Specifically, we verified the translations for terms such as "Stochastic Mixture Kernel," "Pareto Improvement," and "Hierarchical Initialization Strategy" to maintain high academic standards. We also used these tools to check for grammatical consistency in the abstract and conclusion sections.